

Continual Learning and Forgetting

DATA 443/643 · Week 8, note 8.2

Week 8

American University

Ahmad Mousavi

Ideas from Stanford CS224N 2025 L11 (a full Δ is another copy of W) plus the XOR-then-AND lab. Original slides; that course is not copied.

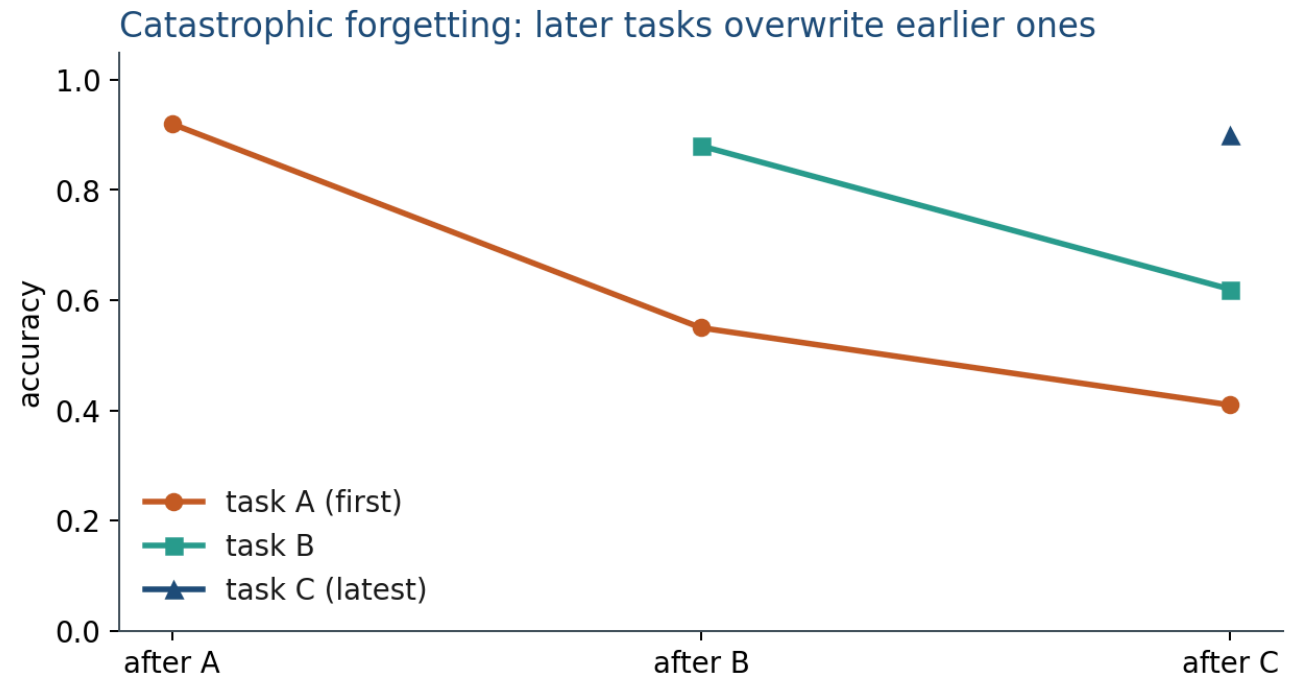
Lecture plan

- 1 Task B walks off task A
- 2 Measure A while you train B
- 3 Replay still moves W
- 4 Freeze W, train an adapter

Key goal: forgetting is A vs B, not train vs test on one task.

A shared W cannot sit at two minima

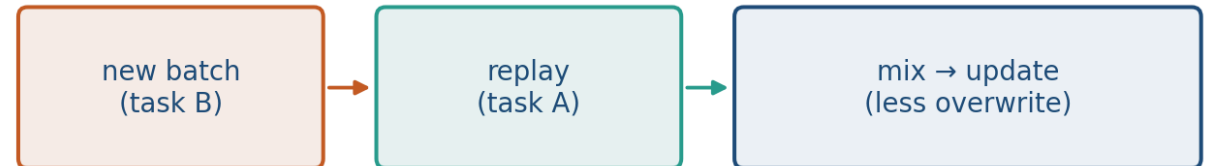
- XOR then AND on the same hidden layer is Lab 8.
- In an LLM the old task is code, languages, refusal style.
- If you only report B, you did not look.



Replay keeps a component of A

- Mix A rows into B batches, or sample the old model.
- 50/50 is the simplest buffer.
- Replay still updates W. That is not LoRA.

Replay (or freeze, or LoRA) is how you keep task A alive



Does W move?

Replay / EWC

Yes. You still need A data (or a Fisher sketch).

Frozen LoRA

No. XOR accuracy on the frozen net cannot move. That is why PEFT exists.

What to take from this lecture

Plot A

Held-out A every k steps of B.

Name it

Overfitting is one task. Forgetting is two.

Lab 8

XOR collapses toward AND unless you replay or freeze.